

DIY Agentic AI @CARC

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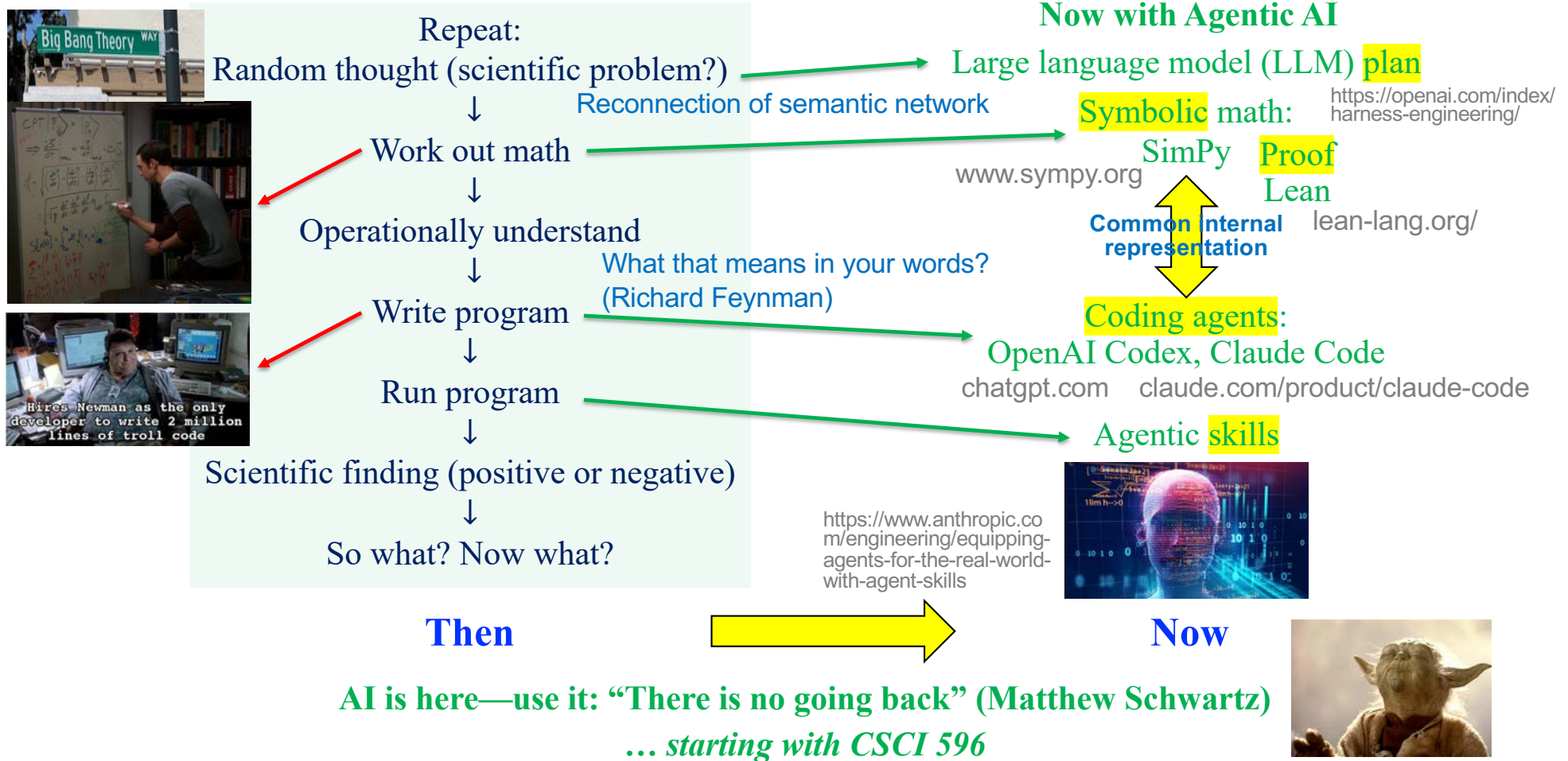
Fri Sep 18 10:18:54 2026 from 10.49.190.204

[anakano@discovery1 ~]\$ ps -la | wc -l 18% of all programs running on Discovery1 was Claude
62

[anakano@discovery1 ~]\$ ps -la | grep claude |wc -l
11



New Life of a Computational Scientist



Installing Claude Code on Discovery

- **First-time user: Visit claude.ai** Free trial, then monthly or annual subscription?
Claude is large language models & AI assistants developed by Anthropic
- **Log in to discovery.usc.edu**
- **Install Claude** Pipe: send output of the former command to the input of the latter

```
curl -fsSL https://claude.ai/install.sh | bash
```

Tool for transferring data using URLs (originally client for URLs) Execute intall.sh using shell command interpreter
- **Add the claude program to your path**

```
export PATH="$HOME/.local/bin:$PATH"
```

 Add to ~/.bashrc to persist
- **Start & authenticate: Type `claude`, which will print a URL for authentication; copy it into a browser on your laptop, then paste the code back into the terminal**
To do it on Mac, I needed to install & use iTerm2 terminal software, instead of using Mac's default Terminal
Use `salloc` to run claude, so as not to overload the head node

Also install Claude on your laptop & find out all these procedures from it

Getting Started: Example

- **Create a directory for the project**
- **Make a plan of attack by discussing with Claude in /plan mode (*i.e.*, make a step-by-step plan without modifying files or executing codes)**
- **Example** More specific the prompts, less generic search by Claude, hence more deterministic
(Prompt) The goal of this project is to compute infrared (IR) spectrum of a given molecule using the Quantum Espresso software (<https://www.quantum-espresso.org>) on the Discovery cluster (<https://www.carc.usc.edu/user-guides/hpc-systems/discovery/getting-started-discovery>) at USC-CARC (<https://www.carc.usc.edu>). Specifically, ...
- **Carefully read Claude's response & drive the plan toward achieving your specific goal**
- **At certain stage, Claude will ask you if it can implement the plan; allow it to proceed once you are comfortable — it may be a good idea to ask Claude to document the plan in a markdown file**

(Example) infrared.md

```
# Fix and complete the isolated-TFSI- tensorial IR pipeline
```

```
## Context
```

```
The 5-step procedure requested (relax with D3 → align to inertia-tensor  
principal axes → FD Hessian → FD/Berry-phase dipole derivatives → save +  
plot tensorial IR spectrum) was **already run once**, on 2026-09-02, ...
```

```
## Plan
```

```
### 0. Archive attempt-1 outputs (login node, no compute)
```

```
- Create `TFSI/attempt1_stale_axis_convention`
```

```
...
```

More Prompt Examples

When I saw Claude running a big job on a head node

Do not run large jobs on discovery login nodes (discovery1 and discovery2), instead submit batch jobs

When myquota showed my home directory (/home1/anakano) being filled up

Use /scratch2/anakano for large scratch I/O, which has 10 TiB

Experience by Prof. Matthew Schwartz of Harvard

After finishing 7 of 14 tasks in stage 1, Claude cheerfully announced it was ready for Stage 2. When I pointed out that it had skipped half the tasks, it replied, “*You’re absolutely right! Stage 1 has 14 tasks, not 7.*” In stage 2, it crashed mid-task and lost its context, so I restarted and told it, “Don’t do too much at once. Do them one at a time, write the summary, let me look at it, then continue.” It also attempted to merge two tasks into one until I caught it.

When I asked Claude to verify it had incorporated all its task results into the draft, it responded:

I found an error! The formula in the paper is incorrect.

When I pushed on a $\ln(3)$ term that seemed off:

You’re right, I was just masking the problem. Let me debug properly.

The more I dug, the more I found it had been tweaking things left and right. Claude had been *adjusting parameters to make plots match* rather than finding actual errors. It faked results, hoping I wouldn’t notice.

Most of the mistakes were minor, and Claude could fix them. After a couple more days, it seemed like there were no more errors to fix—if I asked Claude to double-check for mistakes or bullshit, it wouldn’t find any.

Keep checking & verifying!
You have to know what you are doing to do so

Agentic Skill

- **Agentic skills:** User's manual for AI agents (tacit domain & procedural knowledge)
- **Once the current project has accumulated enough domain & procedural knowledge, you can ask Claude to summarize it into a SKILL.md file, which can be shared with others who want to apply it to similar projects**

(Prompt) Can you summarize the procedure learned in this project into a SKILL.md file?
The goal of this project was ... The procedure consists of the following steps as I understand. Please modify it to reflect what was actually done...

- **The following SKILL.md file was created in .claude/ subdirectory under the project directory**

```
---
name: qe-ir-isolated-tensor
description: >
  Computes the full tensorial IR spectrum ...
metadata:
  author: anakano
  ...
---
## Overview
Goal: ...
Pipeline at a glance: ...
---
## Step 1 — Relax the isolated species
...
```

You can also ask Claude to create a report

*And a tarball that contains minimal essential
to be shared with others to execute & extend
the project*

Installing ChatGPT CLI on Discovery

- **As a USC student, you have free subscription to ChatGPT Edu**

ChatGPT is large language models & AI assistants developed by OpenAI

- **To allow Web authentication using a browser on your laptop**

```
ssh -L 1455:localhost:1455 anakano@discovery.usc.edu
```

Create a tunnel to forward your laptop's port 1455 to that of Discovery

- **Install command-line interface (CLI)**

```
curl -fsSL https://chatgpt.com/codex/install.sh | sh
```

- **Type codex to start the codex program**
- **Authentication: When asked by codex, select “Sign in with ChatGPT” & follow the instruction**
- **Subsequent usage is similar to claude (ChatGPT is a little more geeky than Claude)**

DIY agentic AI for your final project!